DGMS 2019 GEOLOGY • QUESTION 43

Plagioclase Subsolidus Phase Relations: Labradorescence & The Bøggild Gap

Topic: Mineral Physics, Feldspar Petrology & Optics | Primary Reference: Dexter Perkins (Mineralogy, 3rd ed., Ch. 17) & Winter (Petrology, Ch. 6)

Question Statement

Q.43. 'Labradorescence' observed in plagioclase is caused by

(a) Polymorphism
(b) The presence of closely spaced lamellae of An_1 and An_{10} composition
(c) Fine exsolution lamellae of An_{50} to An_{60} composition (Bøggild gap)
(d) Peristerite gap

a Polymorphism

b The presence of closely spaced lamellae of An_1 and An_{10} composition

c **Fine exsolution lamellae of An_{50} to An_{60} composition (Correct Answer)**

d Peristerite gap

✓ Correct Answer: (c) Fine exsolution lamellae of An_{50} to An_{60} composition [Labradorescence is caused by optical interference of visible light from submicroscopic exsolution lamellae within the Bøggild miscibility gap in intermediate plagioclase (An_{48}–An_{58} / An_{50}–An_{60})].

Subsolidus Phase Diagram: Plagioclase Solvi

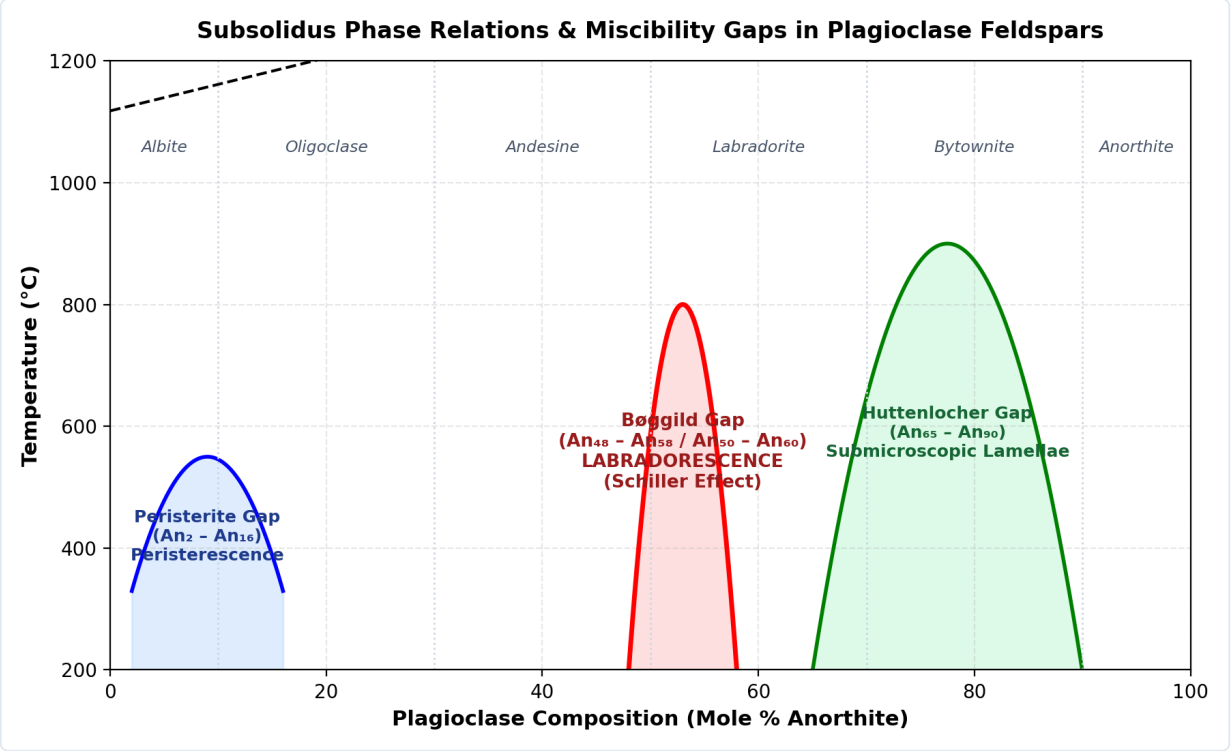


Figure 43.1: Subsolidus Phase Relations and Miscibility Gaps in Plagioclase Feldspars. Plagioclase forms a complete solid solution at magmatic temperatures (>1100°C), but at lower temperatures (subsolidus), three distinct miscibility gaps emerge: (1) **Peristerite gap** ($An_2–An_{16}$), causing peristerescence in albite-oligoclase; (2) **Bøggild gap** ($An_{48}–An_{58}$ / $An_{50}–An_{60}$), where exsolution lamellae on the scale of visible light wavelengths produce iridescent **labradorescence**; and (3) **Huttenlocher gap** ($An_{65}–An_{90}$), occurring in bytownite.

> *"Plagioclases in the range An_{48} to An_{58} (or $\text{An}_{50}–\text{An}_{60}$) unmix upon slow cooling across the* **Bøggild miscibility gap** *into alternating submicroscopic lamellae of slightly differing anorthite contents. The lamellar spacing typically ranges between 100 nm and 300 nm, which matches the wavelength of visible light. Light reflected from these alternating layers undergoes constructive interference, producing the spectacular iridescent blue, green, and golden flashes known as* **labradorescence** *(the Schiller effect)."*
>
> **— Dexter Perkins, *Mineralogy*, Chapter 17 (Tectosilicates: Feldspars), pp. 381–383**

Comprehensive Option Analysis

(c) Fine exsolution lamellae of An_{50} to An_{60} composition (Bøggild Intergrowths) **[CORRECT]**

Optical & Crystallographic Mechanism: Labradorescence is not a pigmentary or chemical color; it is a physical **interference phenomenon (structural color)**:

- As high-temperature plagioclase of labradorite composition ($An_{50}–An_{60}$) cools below ~800°C, it enters the **Bøggild miscibility gap**.
- The crystal unmixes via spinodal decomposition into alternating, planar lamellae differing by only ~10–12% An content (e.g., An_{44} and An_{56}).
- Because these alternating layers have slightly different refractive indices and thicknesses matching the wavelength of visible light ($\lambda \approx 400\text{–}700\text{ nm}$), light reflected from successive lamellar interfaces interferes constructively (Bragg-type diffraction of light). This creates the brilliant spectral Schiller effect known as **labradorescence**.

(a) Polymorphism **[INCORRECT]**

Polymorphism describes a substance with a fixed chemical composition existing in multiple crystal structures (e.g., graphite vs. diamond, calcite vs. aragonite). Plagioclase feldspars are a solid solution series ($\text{NaAlSi}_3\text{O}_8 - \text{CaAl}_2\text{Si}_2\text{O}_8$), not polymorphs. Labradorescence is caused by subsolidus exsolution, not a polymorphic phase transition.

(b) The presence of closely spaced lamellae of An_1 and An_{10} composition **[INCORRECT]**

This composition range corresponds to the albite/peristerite region, not labradorite. Furthermore, lamellae of An_1 and An_{10} do not produce labradorescence; they produce the faint bluish-white sheen of moonstone (peristerescence).

(d) Peristerite gap **[INCORRECT / DISTINCT PHENOMENON]**

The *peristerite gap* is located at the sodic end of the plagioclase series, between **An_2 and An_{16}** (albite and oligoclase). It causes a pale bluish-white, cloudy opalescence termed **peristerescence** (from Greek *peristera*, pigeon, referring to the sheen on a pigeon's neck feathers). It is completely distinct from the vivid, colorful labradorescence of the Bøggild gap ($An_{50}–An_{60}$).

The Three Subsolidus Miscibility Gaps in Plagioclase

Miscibility Gap Name	Composition Range	Plagioclase Variety	Optical Effect Produced	Peak Solvus Temperature
Peristerite Gap	$\text{An}_2 - \text{An}_{16}$	Albite – Oligoclase	**Peristerescence** (Whitish-blue opalescent sheen / moonstone)	~550°C
Bøggild Gap	**$\text{An}_{48} - \text{An}_{58}$ (or $\text{An}_{50} - \text{An}_{60}$)**	**Labradorite**	**Labradorescence** (Spectacular blue, green, golden Schiller effect)	**~800°C**
Huttenlocher Gap	$\text{An}_{65} - \text{An}_{90}$	Bytownite	No visible iridescence (Lamellae are submicroscopic / coarse)	~900°C

Key Takeaway & Exam Strategy

Essential Plagioclase Gap Rules for Competitive Exams:

- **Bøggild Gap $\rightarrow$ $An_{50}–An_{60}$ $\rightarrow$ Labradorescence** (Schiller effect in Labradorite).
- **Peristerite Gap $\rightarrow$ $An_2–An_{16}$ $\rightarrow$ Peristerescence** (Bluish sheen in Albite/Oligoclase).
- **Huttenlocher Gap $\rightarrow$ $An_{65}–An_{90}$** in Bytownite.
- Physical cause is always: **Submicroscopic exsolution lamellae whose thickness matches the wavelength of visible light (~100–300 nm)** causing constructive interference.